1. Iteration Forge — Numerical Methods / Iteration
    a. Approximation by change of sign

Example 1

Show that if $f(x) = x^3 - 4x + 5$, then $f(x) = 0$ has a root between $x = -3$ and $x = -2$.
Find the root correct to 1 decimal place.

$f(-3) = -27 + 12 + 5 = -10$

$f(-2) = -8 + 8 + 5 = 5$

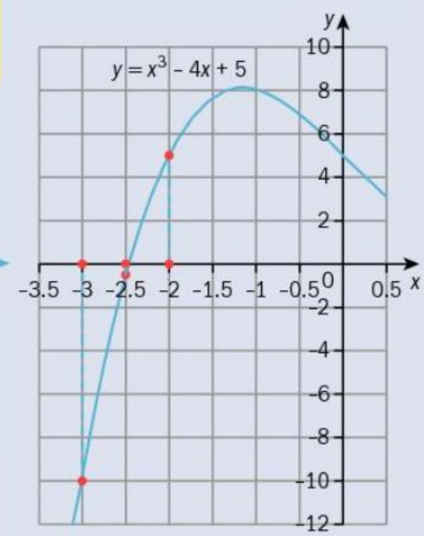
By the sign-change rule, f has a root between $x = -3$ and $x = -2$.

State this explicitly each time.

The next step is to trial a value between –3 and –2. If we take –2.5 then we will know whether the root lies in the interval (–3, –2.5) or in (–2.5, –2).

$f(-2.5) = -0.625 \Rightarrow$ root lies in $(-2.5, -2)$

f(–2.5) = –0.625, which is a very small negative number, and tells us that the root is likely to be close to –2.5. The next trial should be –2.4.



$f(-2.4) = 0.776 \Rightarrow$ root lies in $(-2.5, -2.4)$

The sign of the function at the mid-interval point (i.e. at –2.45) will tell us whether –2.5 or –2.4 is closer to the root.

$f(-2.45) = 0.0938\ldots$ So the root is –2.5 correct to 1 decimal place.

Just looking at the values of the function at –2.5 and –2.4 is not enough. We need to test the sign of the function at –2.45.

Example 2

Show that if $f(x) = x\cos x - \sin x + 1$, then $f(x) = 0$ has a root between $x = 4$ and $x = 5$, where x is in radians, and find the root correct to 2 decimal places.

$f(4) = -0.857\,77...$

$f(5) = 3.377\,23...$ The sign-change rule implies $f(x)$ has a root between $x = 4$ and $x = 5$.

The next step is to trial a value between 4 and 5. Here the function values suggest the root is closer to 4 than to 5, so 4.2 might be sensible for the next trial value.

$f(4.2) = -0.187\,51... \Rightarrow$ root lies in $(4.2, 5)$

Try 4.3:

$f(4.3) = 0.192\,72... \Rightarrow$ root lies in $(4.2, 4.3)$

The sign-change rule implies f has a root between $x = 4.2$ and $x = 4.3$.

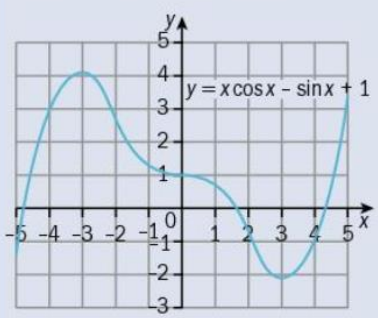
Try 4.25:

$f(4.25) = -0.000\,88... \Rightarrow$ root lies in $(4.25, 4.3)$

$f(4.25)$ is extremely small, which suggests 4.25 is close to the root. The next trial should be 4.26.

$f(4.26) = 0.037\,29... \Rightarrow$ root lies in $(4.25, 4.26)$

$f(4.255) = 0.018\,17...$ so the root is 4.25 correct to 2 decimal places.



The actual root is 4.250 390... Using a spreadsheet or programmable calculator allows roots of equations like this to be obtained to any degree of accuracy available in the program. There are other roots which are shown in the graph on the left.

Example 3

a) Verify by calculation that $\sqrt{(x^3 + 3)} = 2 - x$ has a root between $x = 0$ and $x = 1$.

b) Find the root correct to 1 decimal place.

a) We first need to write the equation in the form $f(x) = 0$ in order to be able to use the sign-change rule, so let $f(x) = x - 2 + \sqrt{(x^3 + 3)}$.

x	$f(x)$	
0	$-0.267\,95...$	The sign-change rule implies there is a root between 0 and 1.
1	1	

Remember to state this.

b)

x	$f(x)$	
0.3	$0.039\,83...$	The root is between 0 and 0.3
0.2	$-0.065\,64...$	The root is between 0.2 and 0.3
0.25	$-0.013\,44...$	The root is between 0.25 and 0.3

A table of values like this, showing the search process with the values, is often the simplest way of keeping track of where we know the root to be located.

The root is 0.3 correct to 1 decimal place.

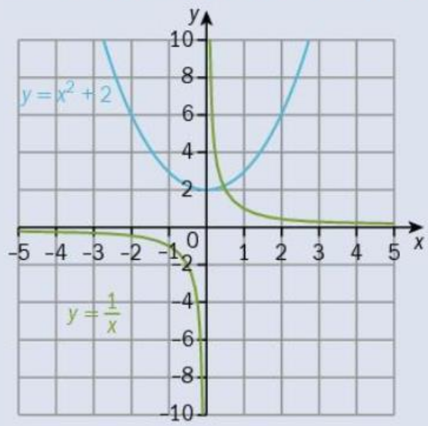
b. Graph proof of root

Example 4

a) Sketch the graphs of $y = x^2 + 2$ and $y = \frac{1}{x}$ on the same set of axes.

b) Using your graphs, explain why there is only one root to the equation $x^3 + 2x - 1 = 0$.

a)



b) The intersection of the graphs $y = x^2 + 2$ and $y = \frac{1}{x}$ is the point at which $x^2 + 2 = \frac{1}{x}$.

Rearranging gives $x^3 + 2x - 1 = 0$.

Since there is only one point of intersection of the two graphs, there is only one root of $x^3 + 2x - 1 = 0$.

c. Find roots using iteration

Example 10

Verify that the equation can be rearranged to $x^2 - 3x - 5 = 0$ in each case.

a) $x = \sqrt{(3x+5)}$ **b)** $x = \frac{3x+5}{x}$ **c)** $x = \frac{5}{x-3}$

a) $x = \sqrt{(3x+5)}$

$\Rightarrow x^2 = 3x + 5$

$\Rightarrow x^2 - 3x - 5 = 0$

b) $x = \frac{3x+5}{x}$

$\Rightarrow x^2 = 3x + 5$

$\Rightarrow x^2 - 3x - 5 = 0$

c) $x = \frac{5}{x-3}$

$\Rightarrow x(x - 3) = 5$

$\Rightarrow x^2 - 3x - 5 = 0$

Example 5

a) Show that the equation $\operatorname{cosec} x = 4 - \frac{1}{2}x^2$ can be rearranged to $x = \sin^{-1}\left(\frac{2}{8 - x^2}\right)$.

b) Use the iterative relationship $x_{n+1} = \sin^{-1}\left(\frac{2}{8 - x_n^{\ 2}}\right)$, with $x_1 = 0$, to find four further approximations x_2, x_3, x_4, x_5, writing down 6 decimal places from your calculator.

c) **Determine** the root of the equation correct to 3 s.f.

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a) $\operatorname{cosec} x = 4 - \frac{1}{2}x^2$

$$\Rightarrow \frac{1}{\sin x} = \frac{8 - x^2}{2}$$

$$\Rightarrow x = \sin^{-1}\left(\frac{2}{8 - x^2}\right)$$

b) $x_1 = 0$

$x_2 = 0.252\,680\ \ldots$

$x_3 = 0.254\,758\ \ldots$

$x_4 = 0.254\,792\ \ldots$

$x_5 = 0.254\,793\ \ldots$

c) The root is 0.255 (to 3 s.f.).

Because the sequence appears to be converging fast, the root is probably 0.254 79 to 5 s.f., but you will not be asked to make judgements like this.

Example 7

a) Show that the equation $xe^x = 1$ has a root between 0 and 1.

b) Show that the equation $xe^x = 1$ can be rearranged into the forms **i)** $x = e^{-x}$ and **ii)** $x = -\ln x$.

c) Use the iterative relationships **i)** $x_{n+1} = e^{-x_n}$ and **ii)** $x_{n+1} = -\ln x_n$, with $x_1 = 0.6$ in both cases, to find eight further approximations $x_2, x_3, x_4, \dots, x_9$, writing down 6 decimal places from your calculator.

a) Writing $xe^x = 1$ as $xe^x - 1 = 0$, let $f(x) = xe^x - 1$.
Then $f(0) = -1$, $f(1) = e - 1 = 1.718\dots$, so by the sign-change rule there is a root between 0 and 1.

b) **i)** $xe^x = 1 \Rightarrow x = \frac{1}{e^x} = e^{-x}$ **ii)** $xe^x = 1 \Rightarrow \ln x + x = 0 \Rightarrow x = -\ln x$

c) **i)** $x_{n+1} = e^{-x_n}$

$x_1 = 0.6$
$x_2 = 0.548\,811\dots$
$x_3 = 0.577\,635\dots$
$x_4 = 0.561\,223\dots$
$x_5 = 0.570\,510\dots$
$x_6 = 0.565\,236\dots$
$x_7 = 0.568\,225\dots$
$x_8 = 0.566\,529\dots$
$x_9 = 0.567\,491\dots$

ii) $x_{n+1} = -\ln x_n$

$x_1 = 0.6$
$x_2 = 0.510\,825$
$x_3 = 0.671\,726\dots$
$x_4 = 0.397\,903\dots$
$x_5 = 0.921\,546\dots$
$x_6 = 0.081\,702\dots$
$x_7 = 2.504\,673\dots$
$x_8 = -0.918\,158\dots$
x_9 gives an error message since we cannot take logs of a negative number.

d. Convergence of iteration

Example 10

Verify that the equation can be rearranged to $x^2 - 3x - 5 = 0$ in each case.

a) $x = \sqrt{(3x+5)}$ **b)** $x = \frac{3x+5}{x}$ **c)** $x = \frac{5}{x-3}$

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c) $x = \frac{5}{x-3}$

$\Rightarrow x(x-3) = 5$
$\Rightarrow x^2 - 3x - 5 = 0$